# Wound Management Procedure

## 1. Scope

| Site | Operational Area | Applicable to |
|---|---|---|
| Armadale Kalamunda Group | All services | Nursing, Midwifery, Medical, Allied Health |

## 2. Purpose

The purpose of this document is to ensure patients who are admitted with, or develop a wound whilst in hospital, will have a comprehensive assessment that encompasses their health status and cultural and environmental factors that may impact their wound healing or risk of developing wounds. As well as to ensure the delivery of optimal patient care and patient safety that is consistent with the requirements of the National Safety and Quality Health Service Standards (second edition).

The multidisciplinary team will develop a plan of care appropriate to the needs of the individual patient while reassessing the clinical status of the patient in response to all interventions and disease processes.

## 3. Medical requirements

- Certain topical products require medical prescription
- A medical order will exist for removal of sutures, staples and drains e.g. post-operative orders, surgeons standing orders, documented verbal orders
- Surgical debridement of wounds will be performed only by a medical practitioner under aseptic conditions
- Infected wounds may require systemic and local treatment
- Routine swabbing of wounds is not recommended unless signs of moderate to severe infection are present. Slough or necrotic tissue is not an indicative for a swab. Where there is doubt about infection, collecting swabs for some clinical presentations is unnecessary i.e. penile discharge, percutaneous endoscopic gastrostomy (PEG) sites, pressure injuries and non-healing ulcers. See AKG Specimen Collection and Pathology Results
- Wound swab should be collected before antimicrobials are commenced.
- For surgical wounds, deep wounds, diabetic foot wounds or if suspected osteomyelitis then sample the most inflamed area.

## 4. Safety and risk documentation

- A Pressure Injury Alert eForm is required for all pressure injuries on admission or sustained during admission
- A Datix CIMS electronic form is required for all hospital acquired pressure injuries and any pressure injury that progresses to the next stage during care

- A Datix CIMS electronic form is required for all non-surgical wounds e.g. skin tear sustained during hospital admission
- All existing wounds and skin conditions to be documented on the *Skin and Pressure Injury Assessment (AKMR 121.9)* within eight (8) hours of admission and appropriate referrals to be made to Dietitian, CNS Wound Care, Occupational Therapist, Podiatrist or others, post assessment

## 5. Nursing management

- Personal protective equipment (PPE) insitu for wound care as clinically indicated
- Perform wound care using asepsis and a non-touch technique. See AKG Aseptic Technique Procedure
- Sterile wound products are single use only
- Document the full wound assessment and commence the *Wound Management Plan* (AKMR 122) (WMP) for each wound
- Photographs are to be utilised, if clinically indicated, to monitor wound progress. Patient consent is to be obtained and documented in medical notes
- Wounds will be reviewed at every dressing change and any changes, concerns or deterioration to be assessed by the multi-disciplinary team, Associate Nurse Unit Manager, Staff Development Nurse or senior nursing staff to evaluate the effectiveness of current treatment and referred if necessary to the CNS Wound Care, Podiatry, surgical service or appropriate tertiary service e.g. plastics, vascular, dermatology, burns, and document in the medical records
- Wound debridement techniques such as Conservative Sharp Wound Debridement require advanced level of skill and competence and should only be undertaken by clinicians who are providing wound care related to their scope of practice e.g. CNS Wound Care, Podiatrist, Medical Officers

## 6. Discharge planning

- Patient's requiring on-going wound management on discharge will require referral to external agencies for wound care e.g. Silverchain, general practitioner; or may require referral to tertiary specialities for further review and management e.g. plastics, vascular, dermatology, burns
- Wound management will be documented on the patient's discharge plan / discharge letter / transfer letter where appropriate. A copy of the Wound Management Plan (AKMR122) must accompany patients during transfer of ward, inter hospital transfer, or discharge to residential care or home with external agencies for wound care.
- Patients discharging with follow up for wound care should have secondary dressings supplied to reinforce appropriately until external agency review
- Inform the patient to report to their GP or service provider looking after their wound if they have:
    - Clinical signs of infection e.g. increased redness, swelling, discharge
    - A deterioration in the wound
    - Any concerns regarding their wound e.g. pain or bleeding

## 7. Procedure

- Involve the patient and carer in wound care and provide wound care and dressing education. Obtain consent prior to assessment and procedure
- Principles of wound management
    - Correct diagnosis and treatment of the underling aetiology of the wound
    - Identify and correct, where possible, factors that may delay wound healing
    - Appropriate wound care
    - Ongoing assessment and management to facilitate wound healing

*Table 1: Factors that may delay wound healing*

| General factors | Local factors |
|---|---|
| Age | Wound management practices |
| Underlying disease | Moisture balance |
| Poor perfusion | Wound temperature and pH |
| Malnutrition | Infection |
| Body mass index extremes | Pressure, friction and shear |
| Disorders of sensation or movement | Foreign bodies |
| Depression, anxiety, fatigue | |
| Radiation therapy | |
| Smoking | |
| Drugs | |

For initial assessment and documentation for each wound, utilise the *Wound Management Plan* (AKMR 122) and document in medical record:

- Anatomical location of the wound
- Wound measurement (length x width x depth)
- Clinical appearance
    - e.g. necrotic, sloughy, granulating, hypergranulating, epithelializing, infected
- Exudate
    - e.g. serous, haemoserous, sanguineous, purulent
- Presence of odour
- Peri-wound

    - e.g. edges rolled or raised, erythema, contact dermatitis, induration, maceration and/or fluctuance, bruised (ecchymosis), oedematous
- Signs of infection
    - e.g. new or increasing pain, erythema, local warmth, swelling, purulent exudate, malodour, abscess, delayed healing, pyrexia, pain
- Pain
    - Pain score to be assessed with each recording of vital signs, and at each dressing change, (more frequently if indicated) using a validated pain scale (e.g. numerical scale). Document findings and management strategy in medical record
- Type
    - e.g. burn, pressure injury, laceration, skin tear, diabetic foot ulcer

### 7.1 Dressing procedure

- Explain procedure to patient and gain consent as appropriate
- Restrict activities around the bed/treatment area e.g. bed making, dusting or mopping floor to reduce environmental cross contamination
- Consider need for analgesia pre-dressing procedure
- Gather equipment and place on bottom shelf of clean trolley
- Position waste bag
- Perform hand hygiene
- Open sterile dressing pack onto top shelf of trolley
- Open and assemble equipment onto sterile field maintaining asepsis
- Don PPE
- Position a waterproof protective sheet under the wound (if required)
- Position dressing pack sterile towel to promote asepsis and protect the surrounding environment
- Carefully remove dressing, expose wound and discard old dressing (Moisten before removal if dressing is sticking to wound bed)
- Remove gloves and perform hand hygiene
- Don clean gloves or sterile gloves as applicable
- Assess the wound and surrounding tissue
- Use aseptic technique throughout procedure
- Use gauze swabs moistened with sterile solution e.g., sodium chloride 0.9%, sterile water or Prontosan solution to cleanse wound

OR

- Gently irrigate the wound (this will depend on the type of wound and assessment)
- Clean and dress wound as per WMP or assessment
- Dispose of waste and PPE and reprocess equipment as applicable
- Perform hand hygiene
- Position and promote patient comfort

- At each dressing change, reassess wound to monitor progress and document in *Wound Progress Notes* (AKMR 122) and medical record

#### 7.1.1 Wound suitability for cleansing with tap water / in the shower

Wounds suitable to be cleansed with tap water/in the shower in a single room with a non-shared ensuite are:

- Surgical wounds healing by primary intention
- Lacerations or grazes
- Chronic wounds with no undermining or underlying structures on view

Wounds NOT suitable to be cleansed with tap water/in the shower in a single room with a non-shared ensuite are:

- Actively bleeding
- Undermined
- If vessel, bone, tendon or underlying structures on view
- If patient is severely immuno-compromised
- Venous access devices, e.g. peripherally-inserted central catheters (PICC), central venous lines, implanted ports

### 7.2 Dressing selection

- No single dressing is perfect for all wounds; a clinician should evaluate individual wounds and choose the best dressing on a case-by-case basis. Wounds must be continually monitored, as their characteristics and dressing requirements change over time
- The acronym TIME (Falanga, 2012) is used to facilitate dressing selection and wound bed preparation for complex and chronic wounds

**T** = Tissue, non-viable
**I** = Infection or inflammation
**M**= Moisture
**E** = Edge of wound, non-advancing or undermined

### 7.3 Tissue

DO NOT DEBRIDE the following:

- Lower extremity wounds in the presence of arterial disease and dry gangrene
- Stable (dry, adherent, intact without erythema or fluctuance) eschar on the heels serves as the body's natural biological cover and should not be removed
- Fungating or ulcerating tumours (cancerous wound)

Debridement methods:

- Surgical
- Mechanical
- Conservative
- Autolytic

Dressings to promote autolytic debridement include:

- Hydrogels e.g. Intrasite Conformable, Intrasite gel
- Cadexomer iodine e.g. Inadine, Iodosorb
- Hydrocolloids e.g. DuoDerm
- Dressing containing a surfactant e.g. Prontosan gel/solution
- Hypertonic saline e.g. Mesalt.

### 7.4 Infection / inflammation

Consider the following dressing:

- Cadexomer iodine e.g. Iodasorb™
- Iodine based products e.g. Inadine™
- Silver based products e.g. Acticoat, Biatain Ag ™
- Hypertonic saline e.g. Mesalt ™
- Prontosan gel/solution

### 7.5 Moisture

- If the wound is too moist it will cause maceration
- If it is too dry (unless that is the goal of care) the wound bed will become desiccated and delay healing

#### 7.5.1 Minimal exudate:

- Hydrogel e.g. Intrasite Gel™
- Calcium alginate e.g. Biatain Alginate, Kaltostat™
- Hydrocolloid e.g. Duoderm CGF™
- Foams e.g. Mepilex Foam™

#### 7.5.2 Moderate to heavy exudate:

- Hydrofibre e.g. Aquacel™
- Speciality absorbent dressings e.g. Zetuvit™
- Negative pressure wound therapy e.g. PICO™, Renasys Touch™
- Hypertonic saline e.g. Mesalt.

### 7.6 Edge of the wound

- May be described as:
    - Undermining (the erosion of tissue along a longitudinal plane below intact skin. Often follows a fascial plane)
    - Hypergranulating (Abnormal granulation tissue that is raised above the level of the peri-wound area. The tissue is soft spongy and friable [bleeds readily])
    - Rolled or raised
    - Wound edges that show evidence of undermining, hypergranulation, or are rolled or raised may require further debridement, skin grafting, or interventions to reduce the bacterial burden

# 8. Burns assessment and management

## 8.1 Burns

Burns are traumatic injuries to the skin and underlying tissues that result from exposure to thermal, chemical, electrical, radiation, cold injury, and friction sources. Classification of the burn and assessment of the depth of injury and extent of total body surface area (TBSA) will guide the resuscitation and ongoing management of the patient including when to transfer to the Burns Unit at Fiona Stanley Hospital (FSH) or Perth Children's Hospital (PCH).

## 8.2 Assessment of burn size and depth

Patient's presenting to the Emergency Department or sustaining burns while admitted will be assessed for the size of the burn and estimated depth. To assess the size, use the Wallace Rule of Nine Chart for adults or the Lund and Browder Chart for Children. Adults with a burn injury of 15% or greater of TBSA and paediatrics with a burn injury of 10% or more signifies a major burn injury and should be transferred to the respective age-appropriate Burns Unit.

Mechanism of injury and patient history should be obtained to assist in predicting the expected depth of the burn injury. Estimate and classify the depth of the burn as either superficial, partial thickness or full thickness.

Burns are categorised as:

- **Superficial-** affect the epidermis. They are very painful, erythematous, may blister, are red beneath the blisters, will heal in 7-10 days
- **Partial thickness-** affect the epidermis and part of the dermis. They are less painful, will be cooler to touch, blisters will occur, and they have a pale mottled appearance and a slower capillary refill. Will heal in two weeks and may scar
- **Full thickness-** affect the epidermal and all the dermal tissue. They are insensate. They can be white, mottled, and avascular, sometimes charred and they will need grafting

Conversion to a deeper burn wound depth is dependent on the following factors:

- Adequate burn first aid provided within the first three hours of injury
- Adequate fluid resuscitation, if clinically indicated, was administered in the immediate resuscitation period (36 hours post injury)
- Presence of infection which may increase the depth of a burn wound
- Appropriate antimicrobial dressings not applied to the burn in the immediate resuscitation period
- Management of the severe burns that may involve the use of certain medications, e.g. inotropes that may affect peripheral circulation and cause the conversion of the burn wound

## 8.3 Adult burns refer to Fiona Stanley Hospital Nursing Practice Standards:

- Burns assessment and initial management refer to FSH Initial Assessment of Minor Burns Guideline

## 8.4 Paediatric burns refer to Perth Children's Hospital Emergency Department (ED) Guidelines:

- Burns assessment and initial management refer to Burns
- Burn analgesia and pain management refer to Burns- Analgesia and dosing

## 9. Diabetic foot ulcers refer to Podiatry

### 9.1 Podiatry service

The Podiatry service focuses on assessment and treatment of adult patients with acute foot problems. All referrals should be made by eReferral. Podiatrists at AHS provide the following:

- Assessment, diagnosis, and management of high-risk foot conditions including foot ulceration, amputation and Charcot neuroarthropathy
- Wound care: including sharp wound debridement, wound dressing plans, shared care with tertiary hospitals
- Neurovascular foot assessment for high-risk foot patients
- Provision of pressure offloading devices
- Footwear assessment, advice and footwear referral for high-risk foot patients and CAEP eligible patients.

Services are primarily limited to people with an active high-risk foot problem and significant medical history such as diabetes, peripheral vascular disease, chronic renal failure, neurological deficit, severe arthritis, immuno-suppressive disorders, and previous history of amputation.

### 9.2 Dressing diabetic foot ulcers

If after-hours and choosing dressings for a diabetic foot ulcer consider using:

- Foam as a primary dressing to manage exudate and not macerate ulcer peri wound
- Be cautious in adding dressings or gels that may macerate peri wound or antimicrobials where no evidence of infection exists
- Consider a sucrose octasulfate impregnated dressing e.g. UrgoStart, in neuro-ischemic ulcers (without severe ischemia) or as an adjunctive treatment in non-infected ulcers that fail to heal after 4-6 weeks despite optimal clinical care

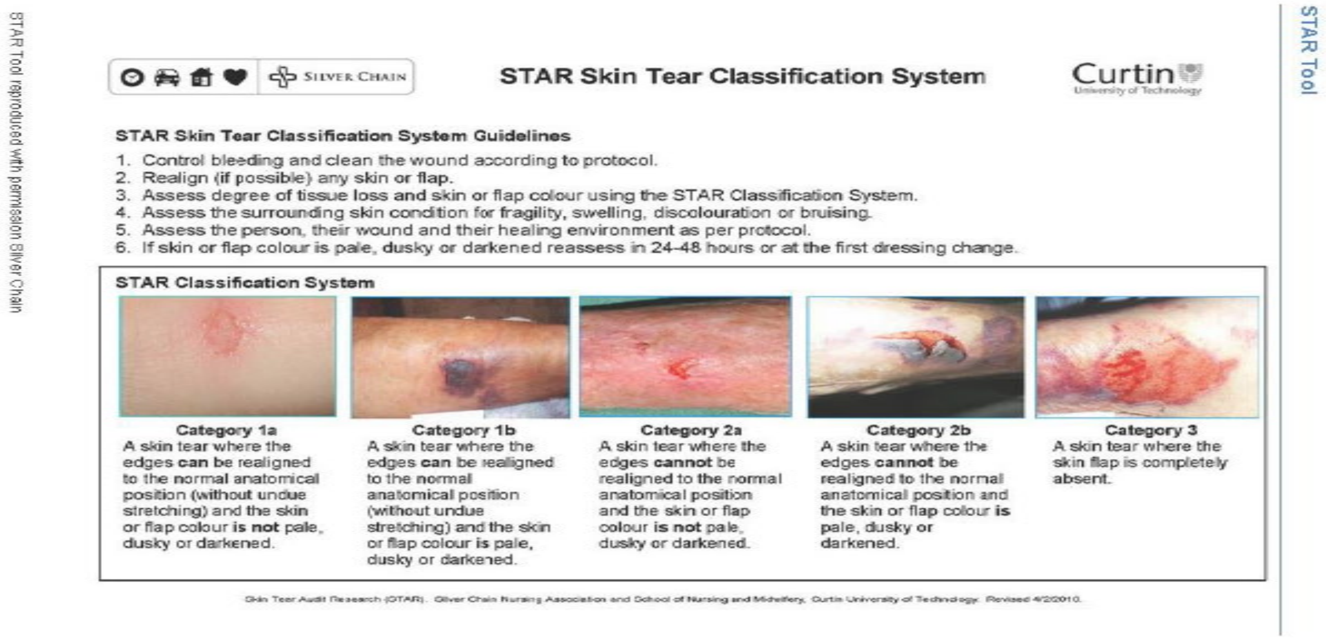
## 10. Skin tear prevention and management

### 10.1 Skin tear classification

A traumatic wound occurring principally on the extremities of older adults. Typically, they result from friction alone or shearing and friction forces.

- Categorise the skin tear using the STAR Skin Tear Classification system.

*Figure 1: STAR Skin Tear Classification system.*




## 10.2 Skin tear procedure

- All admitted hospital patients must have a skin inspection performed within eight hours of admission
- The assessment results will be documented in the patient health record and appropriate interventions implemented
- If no risk is determined, then routine skin care should be provided
- However, if a risk is determined then the following criteria must be adhered to, to keep the patient from injury:
  - Assure a safe environment.
  - Provide an appropriate skin care routine
  - Protect the patient from injury by minimising the risk of friction and shearing forces when positioning, turning, lifting and transfer

*Table 2: Documentation*

| Time | Documentation required | Where |
|---|---|---|
| On Admission | • Condition of skin<br>• Skin Tear Risk Assessment<br>• Malnutrition Score<br>• Implement Skin Protection Strategies<br>• Any skin characteristics, wounds, abrasions, bruises, or other issues revealed by skin assessment | • Complete Skin and Pressure Injury (PI) Assessment (AKMR121.9)<br>• If skin tear present on admission, document in patient health record and complete Wound Management Plan (AKMR 122) |
| Initial wound presentation | • Description of wound<br>• Dressings utilised<br>• Implement Skin Protection Strategies to prevent further wounds | • If patient sustains skin tear while admitted complete an Electronic Clinical Incident report (Datix CIMS). Reassess and complete Skin and PI Assessment (AKMR121.9) and implement Skin Protection Strategies. Document in Care Plan (AKMR 121A) and patient health record |
| High risk for skin tears | • Skin Tear Risk Assessment<br>• Implement Skin Protection Strategies | • Reassess and complete Skin and PI Assessment (AKMR121.9) and implement Skin Protection Strategies, document in Care Plan (AKMR 121A) and the patient health record |

### 10.3 Management of category 1 and 2 skin tears

- Nursing staff will categorise the skin tear using the STAR Skin Tear Classification System, refer to Figure 1
- Skin tears where possible must be assisted to heal by primary intention. This entails recovering the skin flap if possible and gentle fixating the flap to the underlying tissues
- If healing by primary intention is not possible as the skin flap is absent, the wound must be assisted to heal by secondary intention
- Dressing products must maintain a moist wound environment, manage excess exudate, prevent infection, minimise pain and promote patient comfort, whilst minimising the risk of further trauma
- Treating skin tears within 6 hours of injury increases the chance of the skin flap being viable and decreases the risk of infection
- Using a sterile technique, the nurse will:
    - Control bleeding by applying pressure on the wound. Apply a calcium alginate (e.g. Biatain Alginate) with secondary dressing (e.g. Zetuvit) and bandage once bleeding stops and review in 24-48 hours
    - Clean the wound with normal saline. This includes cleaning under the damaged flap with normal saline soaked gauze or sterile soft cotton tips to remove damaged cells and blood / haematoma
    - Using sterile soft cotton tips, carefully approximate the flap without tearing or applying any undue tension
    - For non-infected skin tears apply:

    - Atraumatic silicone dressing (e.g. Mepitel), absorbent dressing (e.g. Zetuvit) and tubular band or crepe bandage.
    - Or silicone coated foam (e.g. Mepilex Border) making sure to indicate the direction of removal
    - Leave dressing intact for 3 days if the wound is not contaminated and change dressing daily if the wound is contaminated.
  - If skin flap is pale, dusky or darkened (Category 1b or 2b) reassess in 24 to 48 hours. The wound may need to be debrided if devitalised tissue is present. If staff member is not competent with Conservative Sharp Wound Debridement, they are to contact the CNS Wound Care or their Nurse Unit Manager (NUM) / Staff Development Nurse (SDN)
  - Document findings and intervention on wound management plan and in the patient's health record
  - Complete a DATIX Clinical Incident Management System (CIMS) form to report injury if injury occurred during hospitalisation
  - Urgent Medical review for, uncontrolled or excessive bleeding, extensive tissue loss and concurrent medical issues.

Note: If there is a skin tear that is difficult to realign, staff are to contact their NUM / SDN or the CNS Wound Care for assistance.

### 10.4 Management of Category 3 skin tears

- Skin tears where the skin flap is completely absent must heal by secondary intention
- The aim of intervention is to promote a moist wound healing environment whilst managing excess exudate
- Dressing options include silicone foams, alginates, and hydrofibres dependent upon exudate levels
- If fixation is required, dressings should be secured with crepe bandage or tubular products rather than steri-strips, fixation sheets, film dressings, hydrocolloid dressings or transpore tape
- Wound exudate will be moderate to high following initial trauma as the wound will be in the inflammatory stage of wound healing. As the wound moves to the proliferative stage of wound healing the exudate level will drop, dressing selection needs to be adjusted accordingly

## 11. Negative pressure wound therapy procedure

### 11.1 Action

Negative Pressure Wound Therapy (NPWT) is a therapeutic technique using topical sub-atmospheric pressure in a vacuum assisted dressing closure to promote healing in acute or chronic wounds and enhance interstitial wound drainage, reduction of oedema, promoting wound contraction, deposition of granulation tissue and reduction of bacterial colonisation.

### 11.2 Contraindications

The use of NPWT is contraindicated in the presence of:

- Necrotic tissue with eschar

- Untreated osteomyelitis
- Malignancy in wound (with exception of palliative care to enhance quality of life)
- Exposed arteries, veins, organs, or nerves
- Non-enteric and unexplored fistulas
- Exposed anastomotic sites

### 11.3 Procedure

- Negative Pressure Wound Therapy (NPWT) dressing ordered by the Surgeon/Consultant, and this is documented in the progress notes or the Operation Record (AKMR 95.2)
- Obtain appropriate device as per Negative Pressure Wound Therapy Device Rental Procedure
- Apply NPWT dressing and device as per manufacturer's guidelines and instructions for use
  - For PICO and Renasys Touch use Smith and Nephew resources https://www.smith-nephew.com/en-au/wa-health-npwt
  - For Prevena use KCI Prevena Dressing Instructions (PDF 3180KB)

## 12. Pressure injuries

For prevention strategies, staging and management of pressure injuries please refer to the Pressure injury Prevention and Management procedure.

## 13. Legislative context

The following documents are required to give effect to this procedure

- HDWA Clinical Handover Policy MP 0095
- HDWA Clinical Incident Management Policy MP 0122/19

## 14. Supporting information

The following documents support the implementation of this procedure:

- Aseptic Technique Procedure AKG-PRO-0045-14
- Clinical Handover
- Clinical Incident Management Procedure AKG-PRO-0087-17
- Pressure Injury Prevention and Management Policy AKG-POL-0375-17
- Specimen Collection and Pathology Results

## 15. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

Organisation-wide data trends and risks associated with pressure injuries and skin tears are to be reported to, and monitored by, the Comprehensive Steering Committee.

## 16. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.11
- Standard 5 – Action 5.3
- Standard 5 – Action 5.7
- Standard 5 – Action 5.10
- Standard 5 – Action 5.12
- Standard 5 – Action 5.21
- Standard 5 – Action 5.22
- Standard 5 – Action 5.23

## 17. References

1. Carville, K. (2017). *Wound Care Manual*. 7th *ed*. Osborne Park, Australia: SilverChain Foundation

## 18. Glossary

The following definitions are relevant to this policy.

| Term | Definition |
|---|---|
| Shearing | Occurs when the skeleton and deep fascia slide downwards with gravity, while the skin and upper fascia remain in the original position |
| Friction | Occurs when two surfaces move across each other. It often removes the superficial layers of the skin.<br>Friction damage often results from a poor lifting technique |
| Primary intention healing | Occurs when there is minimal tissue loss and the edges of the wound are held in close apposition by sutures, clips, tape or dressing products |
| Secondary intention healing | Wounds that heal by secondary intention heal from the base up.<br>The wound edges come together through the formation of granulation tissue and wound contraction, and epithelialisation seals the wound. |

## 19. Policy owner (relating to these procedures)

**Director Nursing and Midwifery**

Enquiries relating to this procedure may be directed to:

Title: Clinical Nurse Specialist Stoma Wound Care
Department: Surgical Services
Email: tariro.magunda@health.wa.gov.au

## 20. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

| Version | Effective from | Effective to | Amendment(s) |
|---|---|---|---|
| V6 | 22/01/2024 | 22/01/2027 | Scheduled review |

## 21. Authorisation

This procedure has been authorised and issued by the Director Nursing and Midwifery.

| | |
|---|---|
| **Authorised by** | Helen Fullarton – Director Nursing and |
| **Authorisation date** | 19/01/2024 |
| **Published date** | 22/01/2024 |